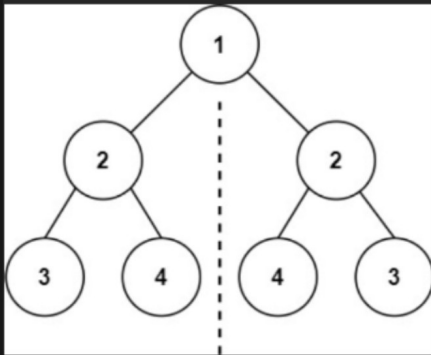


Day 25 :- Tree

LeetCode Pbm No :- 101

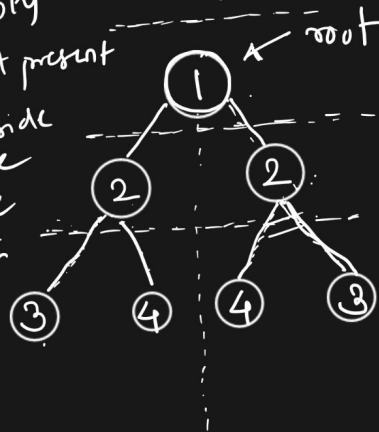
Symmetric Tree



left data right

null	1	not null
------	---	----------

- 1) tree empty
- 2) only root present
- 3) only one side node
- 4) both side node with similar value



in isMirror

root.left $\Rightarrow$ temp₁
root.right $\Rightarrow$ temp₂

class Solution

```
{
    public boolean isSymmetric
        (TreeNode root)
    {
```

```
        if (root == null)
        { return true; }
```

```
        return isMirror(root.left, root.right)
    }
```

```
    public boolean isMirror(TreeNode temp1,
        TreeNode temp2)
```

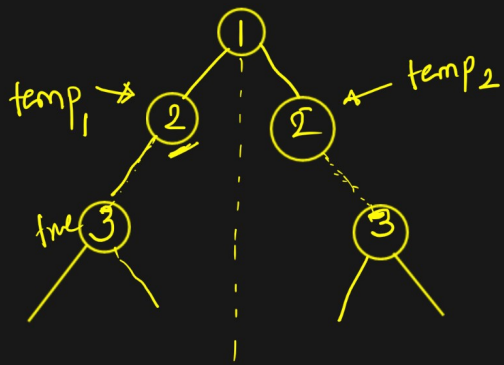
```
    {
        if (temp1 == null && temp2 == null)
            return true;
        if (temp1 == null || temp2 == null)
            return false;
    }
```

return (temp₁.data == temp₂.data) &&

isMirror(temp₁.left, temp₂.right) &&

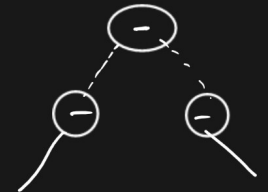
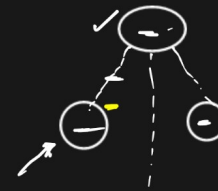
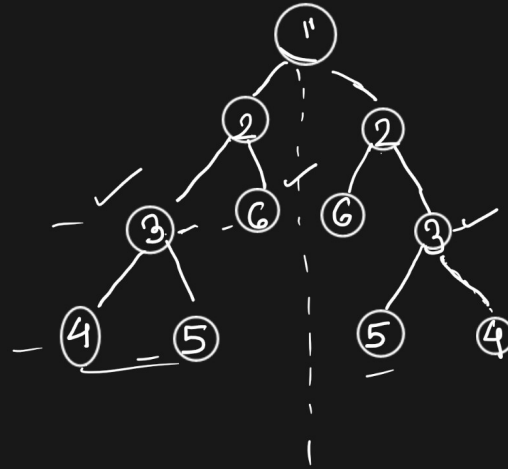
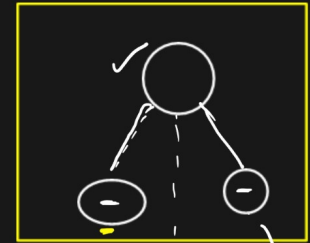
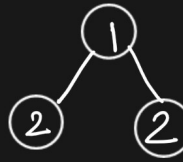
isMirror(temp₁.right, temp₂.left)

}



return (temp₁.data == temp₂.data) &&

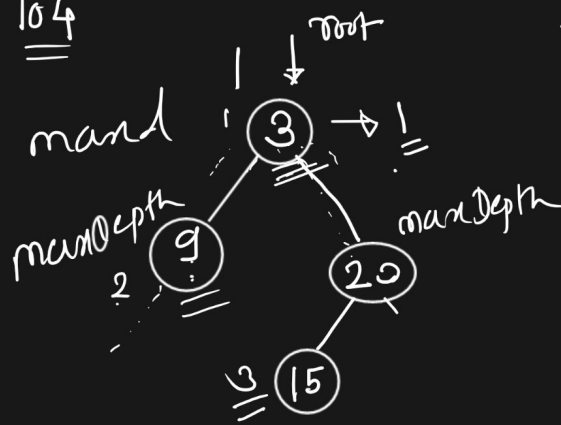
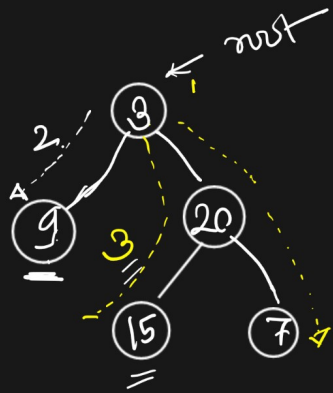
⇒ isMirror(temp₁.left, temp₂.right) &&
isMirror(temp₁.right, temp₂.left)



Traversing

Leetcode pbn No:- 94, 144, 145, 101

Leetcode Pbm No :- 104



```
int maxDepth (TreeNode root)
{
    int result = 0;  $\Rightarrow$  Tree empty
    if (root != null)
```

```

{
    result = result + 1;

    if (root.left == null &&
        root.right != null)
    {
        result = result +
                maxDepth (root.right)
    }

    elseif (root.left != null &&
            root.right == null)
    {
        result = result +
                maxDepth
                (root.left);
    }

    elseif (root.left != null &&
            root.right != null)
    {
        result = result +
                Math.max (maxDepth (root.left),
                          maxDepth (root.right));
    }

    else return result;
}
    
```

maxⁿ depth = no. of nodes
from root to leaf node } maxⁿ depth

$3 \rightarrow 9 \xrightarrow{\text{depth}} 2$
 $3 \rightarrow 15 \xrightarrow{\text{depth}} 3$
 $3 \rightarrow 7 \xrightarrow{\text{depth}} 3$

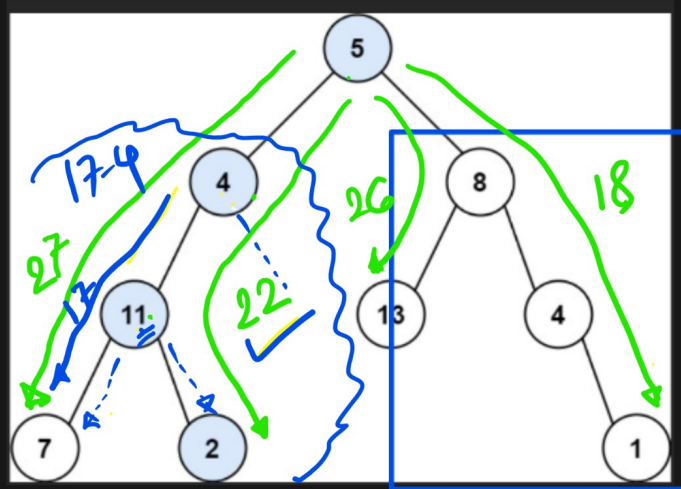
$\left. \begin{matrix} 2 \\ 3 \\ 3 \end{matrix} \right\} \Rightarrow \underline{\underline{3}} \checkmark$

```

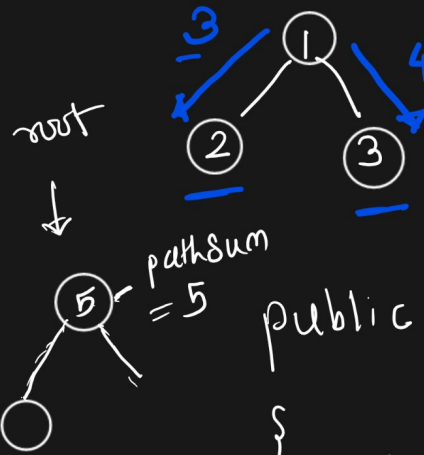
    }
    return result;
}

```

Lectcode Pbm No :- 112 Path of sum



Target sum = 22



Target Sum = 5 $\neq$ 3
 $\neq$ 4

Result = false

```

public boolean pathSum (TreeNode root,
                        int targetSum)
{
    x if (root == null) return false;
    x if (root.left == null & root.right == null)
    {
        return (root.data == targetSum);
    }
    // 22

```

targetSum = targetSum - root.data;

```

return (pathSum(root.left, targetSum) ||
        pathSum(root.right, targetSum))
}

```